

Choosing The Right LED COB



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Why is the LED lighting industry abuzz with light emitting diode chips on boards (LED COBs)? What makes these the current favourite? Which applications are these good for and for which ones should these be avoided? Most importantly, what are the factors based on which you should choose the right LED COB chip? These are some questions I have tried to answer here.

Technologies galore

Before we get down to discussing the parameters based on which you should choose your LED chip—mainly COB LED—let me give you a perspective of the different technologies that power different type of LEDs.

DIP LEDs. Dual inline package (DIP) LED technology, as shown in Fig. 1, is likely what you think of when you picture an LED. It has been around for more than 50 years. Though it is old, DIP LEDs are far from outdated as these are still used extensively for large signs and displays.

These LEDs are typically found in electronic and electrical gadgets

as indicator lights because of their low cost, high brightness and ease of install. DIP LEDs generally produce between three lumens and four lumens per LED.

SMD LED chips. SMD, or surface mounted device, chips (Fig. 2) have become very popular due to their versatility. These are much smaller in comparison to DIP. SMD LEDs have been used to create everything from light bulbs to strip lights.

SMD chips have the ability to put three diodes on the same chip. When a chip includes a red, green and blue diode, you have a chip that can create any colour you want by adjusting the level of output from each individual diode on the chip. Because these are bright and can change colours, these chips are used extensively for LED strip lights and light bulbs.

COB LEDs. COB chips have multiple diodes (more than three) on the same wafer. This is where the similarities between SMD and COB end. In fact, on every COB chip (Fig. 3) there are multiple diodes.

While SMD requires a circuit for every diode included on the chip, a COB device has only one circuit with two contacts for the entire chip, regardless of the number of diodes. This single-circuit design, regardless of the number of diodes on the chip, leads to design simplicity.

More importantly, COB has improved lumen-per-watt ratios in comparison to other LED technologies such as DIP and SMD. Unfortunately, the big drawback of COB (due to single-circuit design) is that multiple channels are necessary to adjust individual levels of light output to create colour-changing effects. Thus, COB lighting is powerful and efficient for single-colour applications, but cannot be used to create colour-changing lights.

Fig. 1: DIP LEDs

